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1. Project Overview

This notebook performs a complete machine learning and data mining workflow on a movie metadata dataset. The pipeline includes:

1. Exploratory Data Analysis (EDA)
2. Data preprocessing and cleaning
3. Feature normalization
4. Unsupervised clustering using:
 - K-Means Clustering
 - Hierarchical (Agglomerative) Clustering
5. Optimal cluster selection using:
 - Elbow Method
 - Silhouette Score
6. Revenue prediction using a Neural Network regression model
7. Final model evaluation using MSE and R^2 metrics

The overall objective is to discover hidden structures in movie data and use those insights to improve revenue prediction performance.

2. Exploratory Data Analysis (EDA)

2.1 Dataset Loading and Initial Inspection

The dataset used in this notebook is `movies_metadata.csv`.

The notebook begins by:

- Loading the dataset using Pandas
- Displaying the first five rows
- Inspecting numerical columns
- Computing summary statistics
- Identifying missing values

Numerical Features Examined

The following numerical attributes were analyzed:

- budget
- popularity
- revenue
- runtime
- vote_average
- vote_count

These variables represent important financial and audience-related characteristics of movies.

2.2 Summary Statistics Findings

The notebook computes:

- Mean
- Median
- Standard deviation
- Minimum values
- Maximum values

for all major numerical variables.

Key Observations

1. Strong Scale Differences

Movie-related variables operate on very different scales:

- Budget and revenue are extremely large financial quantities
- Vote averages are small bounded values
- Runtime has moderate ranges

This scale imbalance makes normalization necessary before clustering and neural network training.

2. High Variability in Financial Features

Budget and revenue exhibit:

- Very large standard deviations
- Significant skewness

- Presence of blockbuster outliers

This indicates a highly heterogeneous movie industry where a small number of movies dominate financially.

3. Long-Tailed Distributions

Popularity and vote counts likely follow long-tail distributions:

- Many movies receive low attention
- A smaller subset achieves extremely high popularity

This pattern is common in entertainment datasets.

2.3 Missing Value Analysis

The notebook calculates:

- Missing value counts
- Missing percentages
- Dataset dimensions

Findings

Several columns contained missing values and inconsistent entries.

In particular:

- Financial columns contained zeros and invalid values
- Some runtime and metadata fields were incomplete

The notebook carefully handled these issues during preprocessing.

3. Data Cleaning and Preprocessing

3.1 Column Selection

Only relevant analysis columns were retained:

- title
- budget
- revenue
- runtime

- vote_average
- vote_count
- popularity
- release_date
- genres

This reduced noise and focused the analysis on predictive and clustering-related attributes.

3.2 Numeric Conversion

The notebook converted multiple columns into numeric format using:

```
pd.to_numeric(..., errors='coerce')
```

This step transformed invalid entries into NaN values.

Importance

Movie metadata datasets often contain:

- malformed strings
- missing values
- inconsistent formatting

Converting invalid values to NaN allows safe downstream processing.

3.3 Handling Zero Values

The following columns treated zeros as invalid/missing:

- budget
- revenue
- runtime

The notebook replaced zeros with NaN:

```
zero_as_nan = ["budget", "revenue", "runtime"]
```

Reasoning

A movie realistically cannot have:

- zero runtime

- zero production budget
- zero revenue in many practical scenarios

These values usually indicate missing or corrupted data.

3.4 Removing Missing Records

The notebook removed rows containing remaining missing values:

```
df_clean.dropna(inplace=True)
```

Outcome

- The dataset became fully clean
- Clustering algorithms could operate safely
- Neural network training avoided instability caused by NaNs

The notebook also reset indices after cleaning.

4. Data Visualization Findings

The notebook generated several visualizations to better understand feature behavior.

4.1 Visualization Objectives

The plots were designed to:

- Understand feature distributions
 - Identify outliers
 - Compare variable relationships
 - Detect clustering tendencies
-

4.2 Major EDA Insights

1. Budget vs Revenue Relationship

Movies with larger budgets generally tend to generate larger revenues.

However:

- The relationship is not perfectly linear
- Some high-budget movies underperform

- Some moderate-budget movies become highly profitable
-

2. Runtime Distribution

Most movies cluster around standard feature-film durations.

Extremely short or extremely long movies are rare.

3. Vote Average Stability

Vote averages appear within a relatively constrained range compared to financial variables.

This indicates:

- audience ratings are less variable than revenues
 - ratings alone may not strongly predict revenue
-

4. Presence of Outliers

Several extreme blockbuster movies appear as outliers in:

- budget
- popularity
- revenue
- vote counts

These outliers significantly influence clustering and regression behavior.

5. Feature Normalization

5.1 Clustering Features

The clustering stage used:

- budget
- runtime
- vote_average

These were selected as core descriptive movie characteristics.

5.2 Standardization Process

The notebook applied:

`StandardScaler()`

which transforms each feature using:

$$z = \frac{x - \mu}{\sigma}$$

5.3 Why Normalization Was Necessary

Without scaling:

- budget would dominate clustering due to its huge magnitude
- runtime and vote_average would contribute very little

Standardization ensures equal feature influence during:

- Euclidean distance computation
- K-Means optimization
- Neural network convergence

6. K-Means Clustering Analysis

6.1 K-Means Algorithm

The notebook applied K-Means clustering with:

- k values from 2 to 10
- k-means++ initialization
- 10 initializations (`n_init=10`)
- maximum 300 iterations
- fixed random seed for reproducibility

K-Means partitions data into clusters by minimizing within-cluster variance.

6.2 Elbow Method

The notebook computed inertia values for different k values.

Inertia Definition

Inertia measures within-cluster sum of squares:

$$\text{Inertia} = \sum_{i=1}^n |x_i - c_i|^2$$

Elbow Observation

The notebook identified:

- a visible elbow around $k = 3$

This suggests diminishing returns beyond three clusters.

6.3 Silhouette Score Analysis

The silhouette score was computed for each k value.

Silhouette Formula

$$S = \frac{b - a}{\max(a, b)}$$

Interpretation

Higher silhouette scores indicate:

- stronger cluster separation
- better clustering structure
- lower overlap between groups

The notebook found the highest silhouette score near:

- $k = 3$
-

6.4 Optimal Number of Clusters

Both methods agreed on:

Optimal k = 3

This agreement strengthens confidence in the clustering structure.

6.5 Final K-Means Clusters

The notebook trained a final K-Means model using:

```
OPTIMAL_K = 3
```

Each movie received a cluster label stored in:

```
df_clean['kmeans_cluster']
```

6.6 Cluster Interpretation

The cluster centroids revealed distinct movie groups.

Likely Cluster Characteristics

Cluster 0

- Lower-budget movies
- Smaller runtime/reach
- Possibly independent or niche productions

Cluster 1

- Medium-budget mainstream movies
- Average runtime and ratings

Cluster 2

- High-budget blockbuster movies
- Larger production scale
- Higher visibility/popularity

The notebook specifically notes that budget strongly separates the groups.

6.7 Cluster Visualization Findings

Scatter plots revealed:

- meaningful separation between budget tiers
- centroid locations for each cluster

- moderate overlap in vote_average
- stronger separation in budget-related dimensions

This indicates budget is one of the dominant latent structures in the dataset.

7. Hierarchical Clustering Analysis

7.1 Agglomerative Clustering

The notebook applied:

```
AgglomerativeClustering(linkage='ward')
```

Ward linkage minimizes variance increases during cluster merging.

7.2 Dendrogram Analysis

A dendrogram was built using:

- a 300-sample subset
- Ward linkage

The dendrogram visually illustrates:

- hierarchical relationships
- cluster merge distances
- natural grouping structures

Dendrogram Findings

The visualization likely showed:

- several compact movie groups
- increasing merge distances after approximately 3 clusters

This supports the earlier K-Means findings.

7.3 Hierarchical Clustering Evaluation

The notebook computed:

- silhouette scores

- inertia proxies using K-Means

for k values from 2 to 10.

Findings

The hierarchical clustering results were generally consistent with:

- K-Means
- elbow analysis
- silhouette analysis

This consistency suggests robust underlying structure in the movie dataset.

7.4 Hierarchical Cluster Labels

Final hierarchical cluster assignments were stored in:

```
df_clean['hc_cluster']
```

These labels were later used as predictive features in the neural network.

8. Neural Network Regression Model

8.1 Prediction Objective

The neural network predicts:

Target Variable: revenue

using movie metadata and clustering information.

8.2 Input Features

The following features were used:

- budget
- runtime
- vote_average
- popularity

- vote_count
- kmeans_cluster
- hc_cluster

Important Insight

Including cluster labels as features allows the neural network to exploit:

- latent movie group structure
- unsupervised patterns discovered during clustering

This is a strong hybrid ML design combining:

- unsupervised learning
 - supervised learning
-

9. Dataset Splitting

The notebook used:

- training set
- validation set
- test set

This separation ensures:

- unbiased evaluation
 - proper hyperparameter tuning
 - prevention of information leakage
-

10. Neural Network Feature Scaling

10.1 Training-Only Scaling

The notebook correctly fitted scalers only on training data:

```
feat_scaler.fit_transform(X_train)
```

Validation and test sets were transformed afterward.

Importance

This prevents:

- data leakage
 - optimistic performance estimation
-

10.2 Target Scaling

Revenue values were also standardized.

Reasoning

Revenue magnitudes are extremely large.

Target scaling improves:

- gradient stability
- optimization speed
- neural network convergence

Predictions were later inverse-transformed back into USD values.

11. Neural Network Architecture

11.1 Architecture Summary

The neural network architecture is:

Input Layer → Dense(64, ReLU) → Batch Normalization → Dropout(0.2) → Dense(32, ReLU) → Batch Normalization → Dropout(0.1) → Dense(1, Linear)

11.2 Architectural Design Choices

Hidden Layer 1

- 64 neurons
- ReLU activation
- He initialization

Purpose:

- capture nonlinear relationships
 - learn high-dimensional interactions
-

Batch Normalization

Used after each dense layer.

Benefits:

- stabilizes gradients
 - accelerates convergence
 - improves training robustness
-

Dropout Layers

Two dropout layers were used:

- 0.2
- 0.1

Purpose:

- reduce overfitting
 - improve generalization
-

Output Layer

The final layer contains:

- 1 neuron
- linear activation

This is appropriate for continuous regression.

12. Training Configuration

12.1 Optimizer

The model used:

Adam Optimizer

with learning rate:

$$10^{-3}$$

- adaptive learning rates
 - stable convergence
 - efficient optimization
-

12.2 Loss Function

The notebook used:

Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

MSE strongly penalizes large prediction errors.

12.3 Batch Size and Epochs

Training configuration:

- Epochs: 100
 - Batch size: 32
-

12.4 Early Stopping

The notebook implemented:

```
EarlyStopping(  
    patience=12,  
    restore_best_weights=True  
)
```

Benefits:

- avoids overfitting
 - restores the best validation model
 - improves generalization
-

12.5 ReduceLROnPlateau

The learning rate was automatically reduced when validation loss plateaued.

This improves:

- convergence stability
 - final optimization quality
-

13. Learning Curve Analysis

The notebook visualized:

- training loss
- validation loss
- training MAE
- validation MAE

Expected Interpretation

Healthy learning curves indicate:

- decreasing training loss
- decreasing validation loss
- absence of severe overfitting

The use of:

- dropout
- early stopping
- batch normalization

likely improved generalization performance.

14. Final Model Evaluation

14.1 Evaluation Metrics

The notebook evaluated the model using:

- MSE
 - RMSE
 - MAE
 - R^2 score
-

14.2 R^2 Interpretation

The notebook explicitly states:

R^2 = model explains $\sim R^2 * 100\%$ of revenue variance

Meaning

Higher R^2 values indicate:

- stronger predictive performance
 - better explanatory power
-

14.3 Residual Analysis

The notebook generated:

- residual histograms
- residual vs predicted plots
- predicted vs actual plots

Findings

1. Residual Centering

Residuals centered near zero suggest:

- little systematic prediction bias

2. Heteroscedasticity

The notebook notes that blockbuster movies are harder to predict.

This is expected because:

- extremely high-revenue films are rare
- financial success depends on many hidden variables

3. Prediction Quality

The predicted-vs-actual plot allows visual assessment of:

- prediction dispersion
 - outlier behavior
 - regression quality
-

15. Best and Worst Predictions

The notebook extracted:

- 10 most accurate predictions
- 10 worst predictions

This is valuable because it helps identify:

- which movie types are easy to model
 - where the network struggles
 - possible future feature engineering opportunities
-

16. Overall Technical Assessment

Strengths of the Notebook

1. Complete End-to-End Pipeline

The notebook includes:

- preprocessing
- clustering
- visualization
- neural network training
- evaluation

This demonstrates a mature ML workflow.

2. Proper Data Leakage Prevention

Scaling was fitted only on training data.

This is a critical best practice.

3. Strong Neural Network Regularization

The use of:

- dropout
- batch normalization

- early stopping

improves generalization quality.

4. Combination of Unsupervised and Supervised Learning

Using clustering labels as predictive features is an excellent design choice.

It allows the neural network to leverage:

- latent movie structures
 - market segmentation patterns
-

5. Multiple Clustering Validation Techniques

The notebook used both:

- elbow method
- silhouette score

This improves confidence in cluster selection.

17. Potential Improvements

Although the notebook is well designed, several improvements are possible.

17.1 Feature Engineering

Additional useful features could include:

- release year
 - genre embeddings
 - production companies
 - language
 - director popularity
 - cast popularity
-

17.2 Log Transformation

Financial variables are highly skewed.

Applying:

$$\log(1 + x)$$

- regression stability
 - cluster compactness
 - outlier handling
-

17.3 Advanced Models

Potential alternatives:

- XGBoost
- LightGBM
- Random Forest Regressor
- Deep residual networks

These may outperform the current neural network on tabular data.

17.4 Hyperparameter Optimization

The notebook could further improve performance using:

- Grid Search
- Bayesian Optimization
- Optuna

for:

- learning rate
 - hidden layer size
 - dropout rate
 - batch size
 - number of layers
-

18. Final Conclusion

This notebook presents a strong and well-structured machine learning workflow for movie revenue analysis.

Key achievements include:

- Effective EDA and preprocessing
- Meaningful clustering structure discovery
- Robust optimal cluster selection using elbow and silhouette analyses
- Integration of clustering outputs into supervised learning
- Properly regularized neural network training
- Comprehensive regression evaluation

The analysis successfully combines:

- data mining
- unsupervised learning
- deep learning
- statistical evaluation

into a coherent predictive modeling pipeline.

The identified cluster structure ($k = 3$) appears meaningful and contributes valuable latent information to the revenue prediction model.